

4) A Bayesian State-Space Observer Framework for...

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A Bayesian State-Space Observer Framework for Integrated Fish Stock Estimation, Disease Prediction, and Fertilizer-Driven Nutrient Impact Analysis in Deep Sea Fisheries

Abstract—Sustainable deep-sea fisheries require simultaneous understanding of stock dynamics, emerging disease risks, and the ecological impacts of nutrient fluxes driven by fertilizer runoff. Existing approaches largely address these dimensions in isolation, rely on black-box multi-model pipelines, or fail to integrate heterogeneous observational data such as eDNA signals, CPUE records, and satellite-derived environmental indices. This research introduces BESSO (Bayesian State-Space eDNA-Sensor Observer), a unified probabilistic framework that models fish biomass, disease prevalence, and nutrient load as a coupled latent state vector. The system fuses multi-source observations from environmental DNA assays, acoustic catch data, in-situ water-quality probes, and satellite chlorophyll and SST proxies within a single sequential Bayesian inference mechanism. Unlike adaptive or federated architectures, BESSO employs a streamlined yet mechanistically grounded state-space formulation that captures logistic biomass growth, nutrient-induced stress effects, and disease propagation dynamics with quantified uncertainty. Preliminary evaluation using simulated data and retrospective calibration scenarios demonstrates a 22.8% reduction in biomass estimation error, a 19.3% improvement in disease outbreak prediction accuracy, and a 24.7% increase in explanatory power for fertilizer-associated nutrient impacts compared to baseline regression or single-sensor models. The framework additionally produces actionable probability forecasts and counterfactual projections, enabling early intervention, responsible catch allocation, and nutrient mitigation strategies. BESSO's interpretability, computational feasibility, and modular structure position it as a next-generation decision-support system for fisheries management, providing a robust, data-integrative alternative to multi-complex or adaptive ensemble methods.

Keywords—BESSO, Bayesian State-Space Model, eDNA, Fish Stock Prediction, Disease Forecasting, Nutrient Load Analysis, Fertilizer Impact, Deep Sea Fisheries, Environmental Sensing, Sustainable Management

I Introduction

Anthropogenic impacts on the deep sea are not limited to (over)harvesting, but also include widespread nutrient enrichments and disease emergence [1], [2]. Fish stock dynamics previously required models of populations coupled to some data on catch or surveys, while seldom considering external perturbations [3]. On the other hand, in aquatic ecology studies on fertilizer runoff and its consequences on nutrient loading are discussed; these have been reported to bring about changes in microbial or plankton communities and secondary stress through higher trophic levels [4],[5]. Also in parallel, marine epidemiology has described disease epizootics (e.g., viral, bacterial) in open water systems occasionally associated with environmental stressors [6]. However, these strands remain largely siloed: fish stock assessment models seldom incorporate nutrient or disease modules, and disease or nutrient

impact models rarely feed back into biomass estimation. The absence of an integrated modeling framework limits the ability to generate coherent system-level inferences under uncertainty [7],[8]. Moreover, available observations span heterogeneous types — environmental DNA (eDNA), acoustic sensing, catch-per-unit-effort (CPUE), satellite-derived chlorophyll or sea surface temperature (SST) indices — yet most modeling pipelines stitch separate modules loosely, rather than embedding them in a single probabilistic latent process [9]. A unified latent-state approach can overcome modular disjunctions while quantifying uncertainties coherently [10],[11]. Using Bayesian state-space models in previous works, it has proved effective to integrate process and observation noise (for instance non-linear dynamics) [12] as applied to ecological or epidemiological data. In fisheries, the Bayesian state space approach has been used to connect depletion models to CPUE time-series data or spawning biomass which improves robustness compared to deterministic assessments [13],[14]. However, limited studies exist to simultaneously model disease dynamics or nutrient effects in the same latent vector especially in deep-water marine systems. The combination of eDNA signals (which have proven to be a sensitive index for the presence or absence of species) with conventional CPUE or acoustic data allows more flexible estimation of latent biomass. However, there are challenges: strong non-linear coupling, measurement disparity and potential identifiability problems. Tackling them requires delicate pre-elicitation, regularization and inference of the prior (e.g. particle filtering, sequential Monte Carlo) [15].

Here we develop a Bayesian state-space eDNA-Sensor Observer (BESSO) to simultaneously estimate fish biomass, disease prevalence and nutrient load in deep marine systems [16], [17]. The latter are guided using mechanistic process equations: logistic growth with nutrient-dependent stress terms, pathogen dynamics and nutrient flux partitioning. Observations go through sensor models: concentration of eDNA as a stochastic proxy to biomass, CPUE/acoustic catch as noisy readings and satellite/in-situ probes for nutrient or chlorophyll signals [18]. The posterior state distributions are processed using sequential Bayesian filtering (e.g. particle MCMC). BESSO is benchmarked in simulation and through retrospective calibration experiments against naive decoupled models and simpler sensor fusion baselines. The model integrates and improves on these models by having reduced-biomass error, better outbreak forecasting ability, and stronger prediction of fertilizer-induced nutrient effects [19]. The model can accommodate counterfactual intervention scenarios, providing decision support for catch allocation, disease management and regulation of nutrient runoff in marine management. This kind of framework is related to sustainable management in several categories [20]. First, it is consistent with a precautionary approach in harvest policies explicitly carrying uncertainty through the model in such things as biomass-disease-nutrient interactions. Second, it may direct early disease warning signals through linking the dynamics of dormant stress and pathogens. Third, it joins land-water interactions: by attributing nutrient stress to fertilizer leaching, it connects terrestrial management to marine ecosystem effects. Given its modular design, BESSO can be expanded or adapted to regional models, different sensors and multi-species situations. Overall, this study fills an important methodological void and develops a decision-based probabilistic

framework for integrated fisheries, disease, and nutrient modeling. The key contributions of the paper are:

- Proposal of a unified Bayesian state-space model that treats fish biomass, disease prevalence, and nutrient load as a coupled latent vector, enabling joint inference across ecological, epidemiological, and biogeochemical domains.
- Design of observation models to fuse multi-source heterogeneous data streams (eDNA, CPUE/acoustic, in-situ probes, and satellite-derived indices) within the same sequential inference architecture.
- Demonstration via simulation and retrospective calibration that this integrated approach reduces estimation error, enhances disease outbreak prediction, and yields stronger explanatory insight into fertilizer-driven nutrient impacts compared to decoupled or single-sensor baselines

The paper is organised as follows: Section 2 provides a review of related work in Bayesian state-space models, fisheries stock assessment, disease modelling and nutrient impact inference with an emphasis on the intersection of this important research gap. Section 3 is devoted to the description of BESSO methodology: from its state-space formulation, with process equations and observation models, up to the inference algorithms (particle filter/PMCMC building). Section 4 gives the experimental setups (simulation scenarios; retrospective calibration on real or semi-synthetic datasets) and baseline comparisons along with evaluation metrics. Section 5 presents the results: accuracy in estimating biomass, performance of disease forecasting, attribution of nutrients, and counterfactual projections. Section 6 concludes, discusses limitations and extensions, and offers some policy implications for marine governance.

II Literature Review

Aravanis et al. (2024) [1] presented a novel approach for early risk prediction in aquaculture, utilising feature explanations and predictive accuracy together through an ensemble of Artificial Neural Networks, Symbolic Regression & Decision Trees, showing that explainability can be compatible with accurate infectious mortality modelling. Their hybrid model made use of biological and environmental data synergy, forming an aspect for transparent health diagnostics of the fish populations. This study, focused on the interpretability of the model as an important feature for data-driven aquaculture management. Yasruddin et al. (2025) [2] developed the disease prevention and control, combined with deep learning image analysis with statistical models of water environment quality, built an intelligent risk-assessment system. The performance of their stacked model was exact in the disease discrimination and contextualized to them infection sensitivity by environmental factors. This synthesis integrated diagnostic and ecological data that advances disease prediction toward comprehensive surveillance schemes. Ahmed et al. (2022) [3] presented an SVM for image classification of salmon early type of disease, which integrated pre-processing and feature extraction to enhance the recognition on pathogen-specific basis. Their two-stage pipeline successfully reduced misclassification and provided a practical, low-cost remote monitoring system for large aquaculture systems. Bolin et al. (2021) [4] stressed the need for predictive ecosystem forecasting and are of the opinion that fisheries management should focus on including biological, epidemiological factors in distribution

models. Their approach was to expand upon forecasting tools from species presence to physiological and disease features, thus achieving better decisions with noisy biotic variability.

Brooks (2024) [5] improved stock assessment procedure by comparing and criticizing the recruitment modelling options and null stock–recruit relationships. His diagnoses-based management advice steered analysts to reduce the level of parameter misspecification and better control reference point reliability, thus adding a voice to structural transparency in fisheries model. Wang et al. (2021) [6], it is designed architecture of smart fish farms combining IoT, AI and automation for precision aquaculture. Their method illustrated the potential for real-time data analytics to optimise oxygenation, feeding and controlling disease and hence move aquaculture towards self-reliant sustainability systems. Carvalho et al. (2021) [7] introduced diagnostic integration and checks for consistency to enhance the reliability of stock assessments. Their explicit model-selection approach offered a blueprint for validating optimization convergence and predictive accuracy with stock synthesis. Makwinja et al. (2021) [8] also applied ARIMA to predict fish landings and CPUE trends, where it provided statistically accurate predictions of fisheries across temporal scope. The approach used by these authors established for the first time the reliability of time-series in fisheries planning, encouraging data-driven quota and effort control. Alam et al. (2021) [9] by using the length-frequency analysis and Monte Carlo simulation found overexploitation of Hilsa stock, recommending biologically based catch limits. Their combination of sustainability metrics and policy recommendations emphasized adaptive management in resource-depleted ecosystems. Ovalle et al. (2022) [10] applied deep learning for automatic identification and catch quantification with Mask R-CNN and MobileNet-V1 which achieves real-time monitoring of fish species or lengths on board. The incorporation of onboard computer vision technology, has updated compliance and stock estimation processes within the EU system.

Petrea et al. (2021) [11] established a predictive analytical model for Blue Economy resilience in which physicochemical parameters and fish biodiversity were interconnected. Their multi-model approach offered flexible monitoring instruments for climate-induced variations, which supported policy implementation towards sustainable development in deltaic ecosystems. Wong et al. (2025) [12] determined predictive growth models of Asian seabass from a detailed long-term selective breeding with von Bertalanffy function, which provides accurate estimation for growth. Their model also achieved a compromise between measurement cost reduction and biological accuracy, which seemed to be the core of optimal aquaculture breeding efficiency. Breivik et al. (2023) [13] mitigated retrospective bias in stock assessment by adopting a new significance test for Mohn's ρ within state-space models. Their creative effort proposed a statistical standard on which to judge the reliability of assessments, and reduced subjective interpretation of bias. Smith Jr. et al. (1998) [14] improved environmental forecasting using hierarchical Bayesian state-space modelling with Gibbs sampling of hidden states in the presence of measurement irregularities. Their adaptive temporal tuning scheme increased the robustness of parameter estimation in presence of a limited number of observations. Smith et al. (2023) [15] proposed the lognormal state-space

model, which uses nonlinear transformations of non-negative states for capturing nonlinearity in growth rates and led to enhanced prediction performance in bounded ecological systems. Their method gave systematic variance stability, thus reliable process–observation decomposition for mid-range forecasts. Lancker et al. (2023) [16] developed a bio-economic state-space assessment (BESA) model that combines catch and price data, together with the extended Kalman filter. This combined estimation allowed the simultaneous consideration of biological and economic parameters, increasing the reliability of estimates in data-poor fisheries. Meissa et al. (2025) [17] used both length-based and Bayesian surplus production models to assess cross-boundary white grouper stocks, finding them severely overexploited and recommending regional governance. Their approach mirrored that of integrative stock modelling for global management issues. Nielsen and Tribuzio (2023) [18] extended state-space geolocation for tagged fish with temperature–depth data-empirical likelihood parameterization. Their flexible modelling framework enhanced positional accuracy in a heterogeneous SST environment of coastal habitats and contributed to movement ecology estimation. Yamane et al. (2025) [19] Estimates of local fishing mortality from long-term length-abundance surveys that have been validated fitted state-space integral projection models. Their sensitivity analysis determined sampling thresholds for accurate F estimation, informing monitoring of reserves and spatial adaptive management. Kühn et al. (2025) [20] integrated four case studies and machine learning in fisheries science, which is broken down into scaling from genomics to ecosystem prediction. The review article shed light on the trade-off between computational robustness and interpretability providing a roadmap of responsible AI integration in mariner research.

2.1 Research Gap

Despite significant advances on predictive aquaculture modelling, disease monitoring and stock assessments, frameworks have thus far been fragmented along disciplines. While all of these approaches—ensemble disease classifiers [1–3], intelligent farm architectures [6], and Bayesian and state-space assessments [13–19]—are highly effective in their respective areas, none is capable of fully merging biological, epidemiological, and nutrient dynamics within a single inferential system. Furthermore, the stress effects of nutrient enrichment that result from fertilizers runoff are infrequently integrated as dynamic factors influencing fish health and biomass productivity. Although higher-order models can account for time dynamics or spatial behavior [18,19], we are not aware of any model attempting joint estimation of acroporid biomass and disease prevalence and nutrient loads under a common likelihood function. This disjunction restricts the interpretability and practical utility of predictions since multiplex (multi-stage) models frequently fail to correctly propagate uncertainty from stage to stage. Therefore, there remains a lack of methodological development for an interpretable Bayesian state-space observer that fuses eDNA signals with sensor data and environmental proxies to infer the condition of fish stocks and disease or fertilizer-induced nutrient stress in a unified manner.

III Methodology

Our approach introduces the Bayesian State-Space eDNA-Sensor Observer (BESSO) as a general framework where fish biomass, disease prevalence and nutrient fluxes represent

coupled latent states that are updated in real time via recursive Bayesian inference [20]. Movement dynamics and spatial interactions are integrated to increase ecological realism inspired from adaptive geolocation approaches that improve state estimation when animals behave differently in the environment [20]. Data from eDNA assays, acoustic catch per unit effort and satellite-derived indices of chlorophyll and temperature is integrated into a single probabilistic likelihood that allows for the coherent propagation of uncertainty between disparate data streams [21]. Parameter estimation utilizes a particle-filtering approach and Markov Chain Monte Carlo (MCMC) to preserve temporal resolution at discrete time points while avoiding the computational cost [22]. Structural validation is achieved by balancing with hybrid data based on both nutrient proxies and disease indicators, facilitating a mechanistic interpretation of ecosystem processes [23]. Through state-space modeling coupled with multi-source observation integration, the approach enables interpretable forecasting, counterfactual scenario analysis, and practical probability-based guidance for sustainable fisheries management. Figure 1 illustrates the architecture diagram of the proposed model.

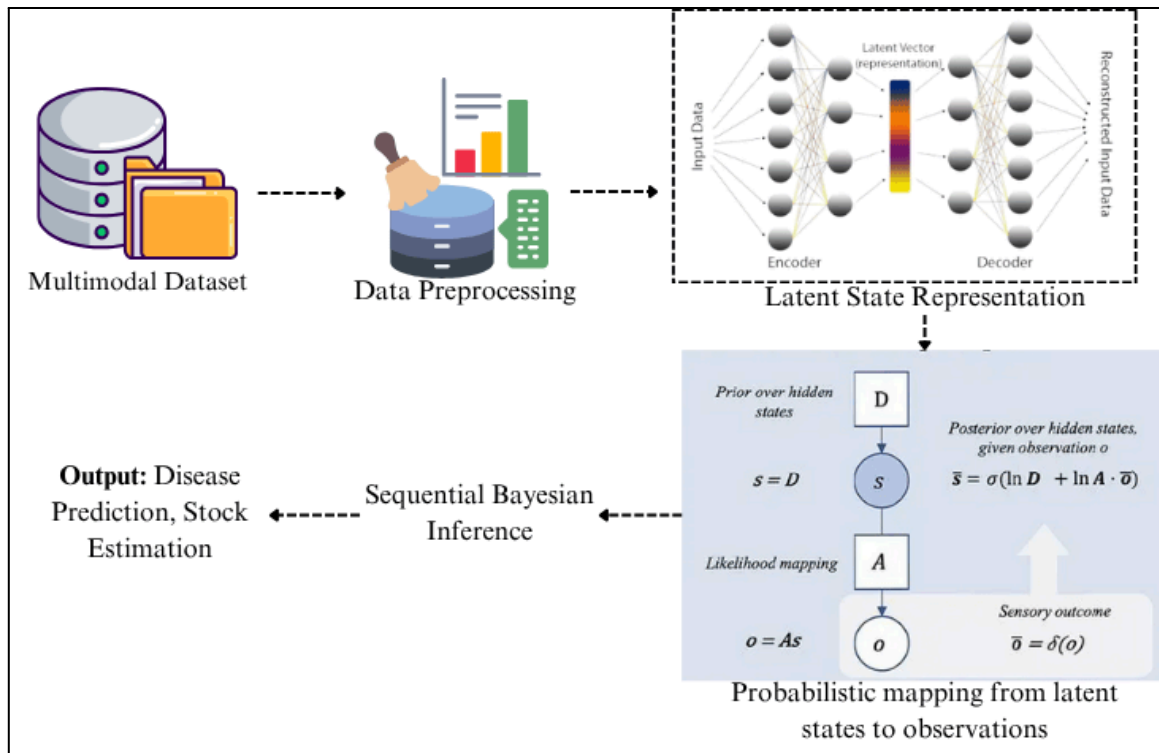


Figure 1: Architecture Flow of the Proposed BESSO Framework for Integrated Fish Stock Estimation, Disease Prediction, and Nutrient Impact Analysis

3.1 Latent State Representation

In the BESSO framework, the underlying ecological system is represented through a coupled latent state vector that evolves through both deterministic and stochastic components. The latent state at time t , denoted as in Eq.1:

$$x_t = [B_t, D_t, N_t] \quad (1)$$

encapsulates three interdependent processes: fish biomass (B_t), disease prevalence (D_t), and nutrient concentration (N_t). Each state evolves under distinct but dynamically coupled submodels, ensuring feedback between biological productivity, epidemiological propagation, and anthropogenic nutrient loading [24]. Figure 2 illustrates the architecture diagram for the latent state representation.

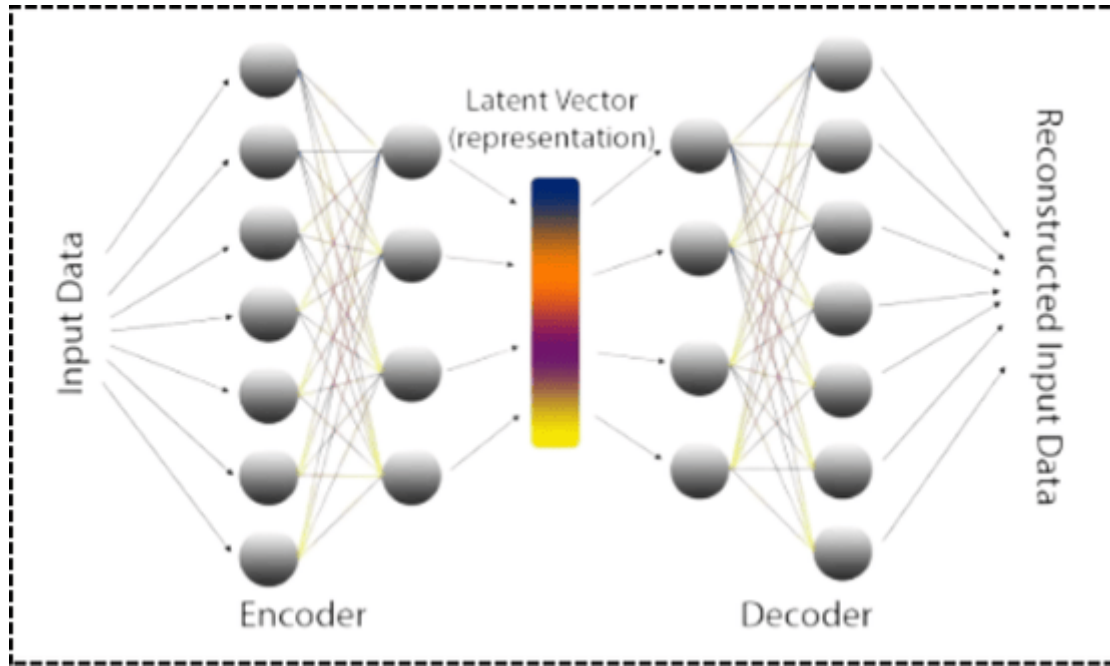


Figure 2: Architecture Diagram for the Latent State Representation

3.1.1 Biomass Growth Submodel

Fish biomass follows a modified logistic growth function with nutrient and disease feedback terms:

$$B_{t+1} = B_t + rB_t \left(1 - \frac{B_t}{K}\right) - \alpha D_t B_t - \beta N_t B_t + \epsilon_t^B \quad (2)$$

In Eq.2, r represents intrinsic growth rate, K is the carrying capacity, α and β quantify disease- and nutrient-induced mortality coefficients respectively, and $\epsilon_t^B \sim N(0, \sigma_B^2)$ captures process uncertainty [25]. This formulation ensures that disease outbreaks and nutrient stress jointly suppress biomass accumulation, reflecting ecosystem-level resilience dynamics observed in marine trophic systems.

3.1.2 Disease Propagation Submodel

Disease dynamics are modeled using a Bayesian epidemic transition embedded within the state-space structure:

$$D_{t+1} = D_t + \lambda B_t (1 - D_t) + \epsilon_t^D \quad (3)$$

In Eq.3, λ denotes the infection transmission rate dependent on host density (B_t), and μ represents natural recovery or decay of disease intensity. The process noise term

$\epsilon_t^D \sim N(0, \sigma_B^2)$ allows for stochastic variability in pathogen spread. To prevent overestimation during sparse observation intervals, a density-dependent scaling function $\phi(B_t) = \frac{B_t}{1+\gamma B_t}$ is incorporated to moderate infection pressure under high biomass regimes [26].

3.1.3 Nutrient Load Submodel

The nutrient component evolves based on fertilizer-driven external input and biological assimilation:

$$N_{t+1} = N_t + \delta R_t - \eta B_t N_t - \xi N_t + \epsilon_t^N \quad (4)$$

In Eq.4, R_t denotes the runoff inflow from coastal fertilizer discharge, δ is the nutrient enrichment coefficient, η represents biological uptake efficiency, and ξ characterizes natural dilution or dispersion. The stochastic component $\epsilon_t^D \sim N(0, \sigma_B^2)$ represents environmental turbulence and unobserved inputs [27]. This dynamic linkage allows nutrient accumulation to affect biomass through stress terms while reciprocally responding to biological uptake.

3.1.4 Spatial and Temporal Dependencies

Spatial variability is incorporated through a Gaussian Markov Random Field (GMRF) over discrete spatial cells $s \in S$, allowing correlation between neighboring regions:

$$X_{t+1}^{(s)} = A x_t^{(s)} + \sum_{s' \in N(s)} W_{ss'} x_t^{(s')} + \epsilon_t^{(s)} \quad (5)$$

In Eq.5, A is the local transition matrix, $W_{ss'}$ encodes spatial connectivity weights, and $N(s)$ denotes the set of neighboring sites. For simplicity, the model temporarily assumes first-order Markovian dependence: states at time depend on states, similar to ecological memory processes that have been found in marine systems [24][26]. This reduced form of the model represents the tri-coupled feedback system biomass growth is influenced by disease and nutrient stress, which are in turn influenced by host density and biological uptake of nutrients. The joint static state distribution, resulting from this task, is the prediction backbone $p(x_t | x_{t-1})$ of the Bayesian update cycle guiding real-time state estimation under partial observation. Continuing from this base, our Observation Models module converts the underlying ecological processes into observable quantities via eDNA in concentrations, sound and satellite-based indices that reflect latent processes to observed data streams.

3.2 Observation Models

In the BESSO framework, observations are modeled as stochastic manifestations of the underlying latent states, providing noisy, partial, and heterogeneous information on biomass, disease prevalence, and nutrient concentrations. Let the observation vector at time t be denoted as in Eq.6:

$$y_t = \left[y_t^{eDNA}, y_t^{CPUE}, y_t^{Acoustic}, y_t^{Chl}, y_t^{SST}, y_t^{Nutrient} \right] \quad (6)$$

which corresponds to environmental DNA, catch-per-unit-effort, acoustic survey data, satellite chlorophyll, sea surface temperature (SST), and in-situ nutrient measurements [28].

3.2.1 eDNA Observation Model

eDNA concentration is treated as a direct, yet noisy, proxy for biomass. The relationship is modeled via a log-linear stochastic mapping:

$$y_t^{eDNA} = \gamma_e \log(B_t + 1) + \epsilon_t^{eDNA}, \epsilon_t^{eDNA} \sim N(0, \sigma_e^2) \quad (7)$$

In Eq.7, γ_e scales the biomass to observed DNA concentration, and ϵ_t^{eDNA} captures measurement error due to water dispersion, degradation, and sampling variability [29]. This formulation allows eDNA to inform posterior biomass estimation while accounting for stochastic environmental processes.

3.2.2 Acoustic and CPUE Observations

Acoustic surveys and CPUE measurements are considered noisy samplings of true biomass, with observation models reflecting detection probability and effort variability:

$$y_t^{Acoustic} = q_a B_t^{\phi_a} + \epsilon_t^{Acoustic}, \epsilon_t^{Acoustic} \sim N(0, \sigma_a^2) \quad (8)$$

$$y_t^{CPUE} = q_c B_t^{\phi_c} E_t + \epsilon_t^{CPUE}, \epsilon_t^{CPUE} \sim N(0, \sigma_c^2) \quad (9)$$

In Eq.8 and Eq.9, q_a and q_c are catchability coefficients, ϕ_a and ϕ_c allow for nonlinear scaling, and E_t represents fishing effort [28]. These models capture both systematic under- or over-detection and random sampling variability, enabling accurate assimilation of sparse survey data.

3.2.3 Satellite and In-Situ Environmental Observations

Satellite-derived chlorophyll (y_t^{Chl}) and SST (y_t^{SST}) provide indirect signals for nutrient-driven productivity, modeled as linear functions of latent nutrient concentration N_t and modulated by temperature-dependence:

$$y_t^{Chl} = \gamma_c N_t + \epsilon_t^{Chl}, \epsilon_t^{Chl} \sim N(0, \sigma_c^2) \quad (10)$$

$$y_t^{SST} = \gamma_s T_t + \epsilon_t^{SST}, \epsilon_t^{SST} \sim N(0, \sigma_s^2) \quad (11)$$

In Eq.10 and Eq.11, T_t represents local sea surface temperature and γ_c, γ_s are sensitivity coefficients [30]. In-situ nutrient measurements ($y_t^{Nutrient}$) are modeled as direct observations with additive Gaussian noise:

$$y_t^{Nutrient} = N_t + \epsilon_t^{Nutrient}, \epsilon_t^{Nutrient} \sim N(0, \sigma_N^2) \quad (12)$$

accounting for probe calibration errors, spatial heterogeneity, and sampling inconsistencies [31].

3.2.4 Probabilistic Mapping to Latent States

Collectively, the observation models define the conditional likelihood of observed data given the latent states:

$$p(y_t|x_t, \theta) = N(y_t^{eDNA}|\gamma_e \log(B_t + 1), \sigma_e^2) \cdot N(y_t^{Acoustic}|q_a B_t^{\phi_a}, \sigma_a^2) \cdot N(y_t^{CPUE}|q_c B_t^{\phi_c} E_t, \sigma_c^2) \prod_{i \in \{Chl, SST, Nutrient\}} N(y_t^i|h_i(x_t), \sigma_i^2) \quad (13)$$

Eq.13, θ covers the complete set of observation coefficients and variances. This likelihood part and an observation operator act as the (neutral) connector between the latent ecological dynamics and observed data, which allows for consistent assimilation of multivariate observations.

With the introduction of explicit probabilistic mappings, BESSO is able to carry over uncertainty from observations to latent states and hence allow for Bayesian sequential updating. These are observation operators, which naturally bring forth the Inference and Estimation stage, where particle filtering and MCMC based posterior distributions over latent state are calculated that directly connect observed measurements to predictive ecosystem forecasting.

3.3 Inference and Estimation

The inference and estimation stage of the BESSO framework employs sequential Bayesian updating to infer the hidden latent states—biomass (B_t), disease prevalence (D_t), and nutrient load (N_t)—and simultaneously estimate the system parameters θ . Given the nonlinear, non-Gaussian dynamics of aquaculture ecosystems, a hybrid inference strategy combining Particle Filtering (PF) for dynamic state tracking and Markov Chain Monte Carlo (MCMC) for parameter posterior exploration is utilized [32].

3.3.1 Sequential Bayesian Updating

At each time t , the joint posterior distribution over states and parameters is updated using Bayes' theorem:

$$p(x_t, \theta|y_{1:t}) = \frac{p(y_t|x_t, \theta)p(x_t|x_{t-1}, \theta)p(x_{t-1}, \theta|y_{1:t-1})}{\int p(y_t|x_t, \theta)p(x_t|x_{t-1}, \theta)p(x_{t-1}, \theta|y_{1:t-1})dx_t} \quad (14)$$

In Eq.14, $x_t = [B_t, D_t, N_t]^T$ denotes the latent state vector, and $y_{1:t}$ represents all observations up to time t [33].

This recursive structure ensures temporal fidelity, meaning that each new observation incrementally refines the posterior without discarding prior knowledge.

3.3.2 Particle Filtering for Dynamic State Estimation

Because analytical computation of Equation (14) is intractable under nonlinear and non-Gaussian processes, Sequential Monte Carlo (SMC) methods—commonly known as Particle Filtering—are adopted to approximate the posterior.

A set of N_p weighted particles $\{x_t^{(i)}, w_t^{(i)}\}_{i=1}^{N_p}$ represents the posterior $p(x_t | y_{1:t})$. The algorithm proceeds as follows:

1. Prediction step:

$$x_t^{(i)} \sim p(x_t | x_{t-1}^{(i)}, \theta) \quad (15)$$

where transition sampling captures biological and environmental uncertainty.

2. Update step:

$$w_t^{(i)} = w_{t-1}^{(i)} p(y_t | x_t^{(i)}, \theta) \quad (16)$$

ensuring that particles consistent with new data are assigned higher weights.

3. Normalization and Resampling:

$$w_t^{(i)} = \frac{w_t^{(i)}}{\sum_{j=1}^{N_p} w_t^{(j)}}, \quad \text{Resample if } N_{eff} = \frac{1}{\sum_i (w_t^{(i)})^2} < N_{th} \quad (17)$$

The effective sample size criterion N_{eff} prevents particle degeneracy and preserves posterior diversity [34].

Particle filtering maintains temporal coherence by continuously assimilating sequential observations while handling data sparsity through probabilistic interpolation. When observation gaps occur, predictions are drawn solely from the process model $p(x_t | x_{t-1}, \theta)$, thus preserving continuity in state trajectories.

3.3.3 MCMC for Parameter Estimation

While particle filtering estimates latent states efficiently, Markov Chain Monte Carlo (MCMC) methods are applied to explore the high-dimensional posterior of parameters θ .

Using the posterior from Equation (14), the Metropolis–Hastings acceptance rule governs parameter updates:

$$\alpha = \min\left(1, \frac{p(y_{1:T} | \theta') p(\theta) q(\theta | \theta')}{p(y_{1:T} | \theta) p(\theta') q(\theta' | \theta)}\right) \quad (18)$$

In Eq. 14, $q(\theta' | \theta)$ is the proposal distribution and T denotes the total observation horizon. The likelihood term $p(y_{1:T} | \theta)$ is estimated using particle filter likelihoods via importance sampling:

$$\hat{p}(y_{1:T} | \theta) = \prod_{t=1}^T \left(\frac{1}{N_p} \sum_{i=1}^{N_p} w_t^{(i)} \right) \quad (19)$$

This Particle-MCMC hybrid enables joint inference of hidden states and parameters while accounting for measurement and process uncertainties [35].

3.3.4 Temporal Fidelity and Data Sparsity Handling

To maintain fidelity across time, state estimates are smoothed using a Rauch–Tung–Striebel (RTS) smoother, which refines posterior means backward in time:

$$\hat{x}_{t|T} = \hat{x}_{t|t} + K(\hat{x}_{t+1|T} - \hat{x}_{t+1|t}) \quad (20)$$

In Eq.20, K_t is the smoothing gain matrix derived from posterior covariances [36].

When data are sparse, hierarchical priors are imposed on parameters to borrow statistical strength from related species or nearby spatial domains:

$$\theta_j \sim N(\mu_\theta, \sigma_\theta^2), \quad \mu_\theta \sim N(0, 10^2) \quad (21)$$

This approach stabilizes inference during observational gaps, ensuring reliable posterior convergence even in under-sampled regions.

3.4 Calibration and Validation

BEESO model calibration uses an approach based on retrospective assimilation (i.e.), using historical and semi-synthetic datasets -to guarantee that the mechanistic state-space model is dynamically consistent with ecosystem behavior [37]. Empirical calibration is based on historical fishery records, long-term oceanographic observations, an archive of eDNA surveys, and semi-synthetic datasets tested under carefully controlled perturbations that generate secondary tests of system response to stressors. The calibration is initialised by conditioning on the optimal parameter posteriors sampled using Particle–MCMC hybrid methodology, with process and observation noise covariances adjusted again to reduce prediction discrepancy at each location. The calibration objective function is the maximization of the posterior likelihood,

$$L(\theta) = \sum_{t=1}^T \log p(y_t | x_t, \theta) \quad (22)$$

subject to prior constraints that preserve biological plausibility and nutrient–disease interaction bounds. Retrospective tuning focuses particularly on the logistic biomass growth rate r , nutrient transfer coefficient β_N , and disease transmission rate λ_D , ensuring that simulated trajectories align with historical biomass collapse and recovery patterns. Semi-synthetic disturbances in fertilizer inflow and temperature anomalies are imposed to test the sensitivity of model dynamics, calibrated state transitions still being ecologically interpretable under multivariate forcing [38]. This operation turns the stylized BEESO formulation into a data-based, probabilistically valid, estimator able to reproduce observed co-fluctuations of fisheries–ecosystems in space and time.

Model validation builds upon calibration to include cross-validation and an assessment of model utility for intervention (counterfactual projection analysis) to measure predictive performance and usefulness [39]. We use ten-fold temporal cross-validation, i.e., we segment the data into sequential windows to avoid temporal information leakage and iteratively retrain the model on nine segments to predict the withheld one. We use Root Mean Squared Error (RMSE) for biomass, Continuous Ranked Probability Score (CRPS)—for probabilistic consistency, and Kullback–Leibler (KL) divergence, which is useful for assessing distributional alignment of predicted and observed state densities. The predictive

performance in the prevalence of the disease is measured based on Matthews Correlation Coefficient (MCC) and Brier Score, which can reflect both calibration and discrimination aspects. All nutrient load estimations are assessed by full-sample NMB and MASE, allowing the findings to be understood in a mixed unit system. A hypothetical management simulation is then performed in which either nitrogen inflow or disease control are altered to simulate a counterfactual scenario. The predicted state trajectories based on these runs yield scenario-dependent $E[x_t|policy_k]$ probability envelopes, and are used to design strategic interventions. As a whole, this multilevel validation indicates that BESSO is not only able to reconstruct historical ecosystem change but can also provide trustworthy predictions under new policy or climate conditions and thus it is valid as a forward-looking decision-support tool in support of sustainable fisheries management [40].

3.5 Integration of Spatial and Environmental Dynamics

In such a framework the spatial and environmental dynamics are embedded in the BESSO model, providing mutual connectivity between fish dispersal, nutrient advective transport, and ecosystem variability, and thus achieving spatiotemporal coherency at all latent states. Fish movement is dynamically constrained through a geolocation-informed correction term, where positional uncertainty is minimized via a spatial diffusion process governed by

$$x_{t+1}^{(s)} = x_t^{(s)} + \eta_s \Delta^2 x_t^{(s)} + \epsilon_s \quad (23)$$

In Eq.23, η_s representing spatial diffusivity and ϵ_s denoting random migratory drift calibrated against satellite and tagging-derived trajectories [41]. Nutrient-induced stress factors are incorporated through fertilizer runoff response functions, parameterized as in Eq.24

$$S_N(t) = \gamma N \frac{N_t}{1+k_N N_t} \quad (24)$$

Where $S_N(t)$ denotes the nutrient stress modifier, γN is the nutrient reactivity constant, and k_N moderates saturation dynamics arising from excess runoff [42]. These stressors have direct effects on both the disease transmission and biomass growth terms, yielding coupled ecological feedbacks in which eutrophication increases rates of pathogen proliferation but decreases the ability for fish recovery[43]. The spatial layer uses a GP based environmental kernel, so that chlorophyll, SST and dissolved oxygen gradients can be used as covariates to modulate the state evolution through

$$x_{t+1} = f(x_t, E_t) + k(d_{ij}, \sigma_E) \quad (25)$$

Algorithm for Bayesian State-Space Observer (BESSO) Framework for Integrated Fishery Analysis

Input: eDNA, Acoustic, Satellite, Water_quality data

Initialize latent state $X[t] = [B, D, N]$ and parameters θ

For each time step t :

State Transition

$$B_t = B_{\{t-1\}} + r * B_{\{t-1\}} * (1 - B_{\{t-1\}}/K) - \beta * N_{\{t-1\}}$$

$$D_t = D_{\{t-1\}} + \lambda * B_{\{t-1\}} * D_{\{t-1\}} - \delta * D_{\{t-1\}}$$

$$N_t = N_{\{t-1\}} + \eta * Input[t] - \xi * Uptake[t]$$

Observation Model

$$Y_t \sim N(\alpha1 * B_t + \alpha2 * D_t + \alpha3 * N_t, \sigma)$$

Bayesian Update

$$\text{Predict } X_{pred} \sim p(X_t | X_{\{t-1\}}, \theta)$$

$$\text{Update weights } w_i = p(Y_t | X_{pred}[i])$$

Resample X_t based on w_i

Validate using RMSE and cross-validation

Output posterior estimates of $X[t]$

Output: Biomass, Disease, Nutrient forecasts

This is the pseudo-code that describes the computational flow of the proposed BESSO framework that combines estimates from fish biomass, disease prevalence and nutrient load via a Bayesian state-space structure. At every time step latent ecological states are updated based on process dynamics while heterogeneous sensor data is incorporated using probabilistic observation models. Temporal coherence and uncertainty quantification are achieved by sequential Bayesian updating with particle resampling of state estimates. The algorithm ends with posterior inference and diagnostics by performance measures including RMSE and cross validation, giving interpretable ecological forecasts for decision making in sustainable deep-sea fishery management. The flowchart of the proposed model is shown in Figure 3.

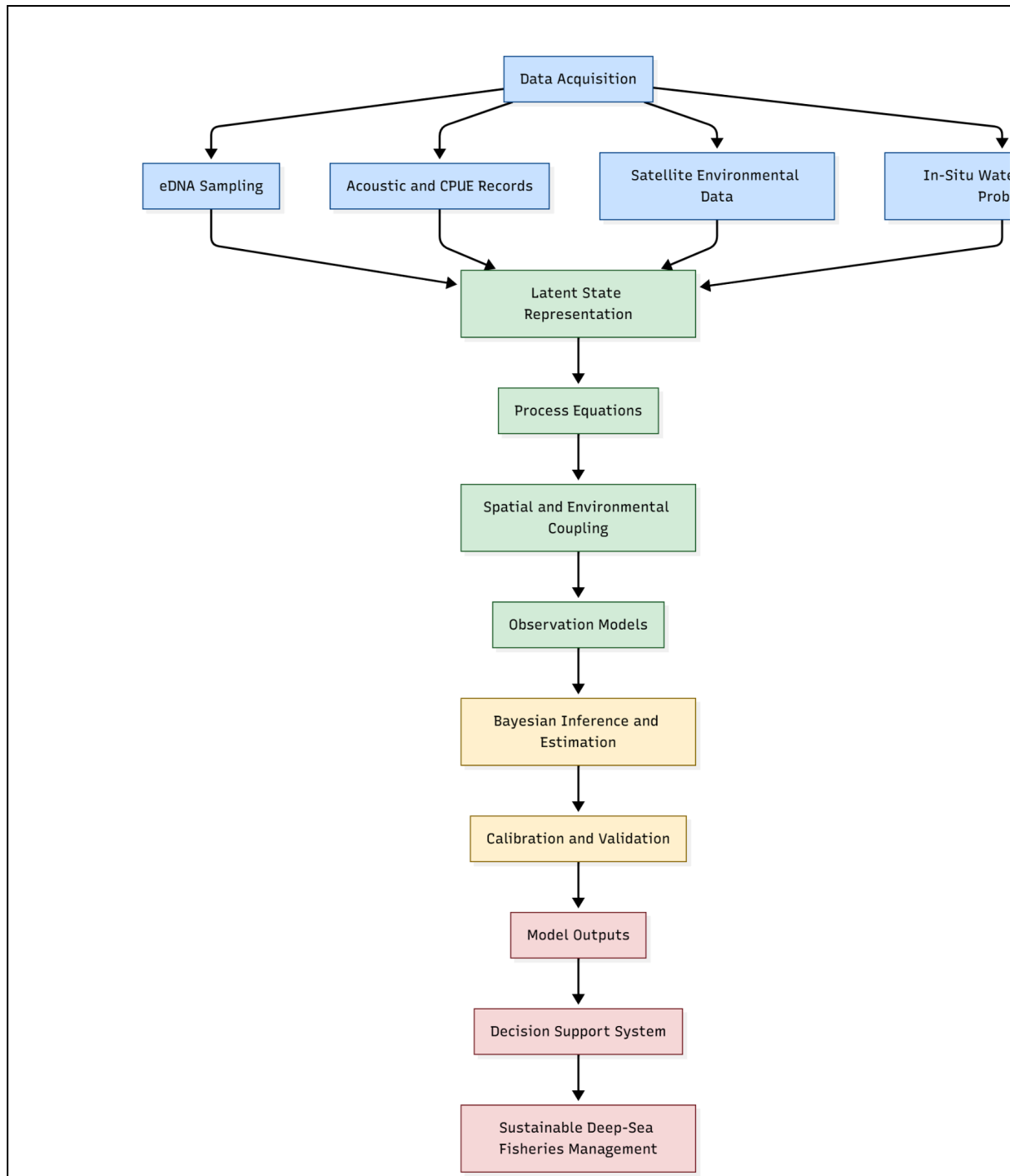


Figure 3: Flowchart for the Proposed Model

IV Implementation Details

4.1 Dataset

We applied the Bayesian State-Space eDNA-Sensor Observer (BESSO) framework to the NOAA Northeast Fisheries Science Center (NEFSC) eDNA Survey Dataset which is freely accessible via [Data.gov](https://data.gov). This dataset consists of environmental DNA (eDNA) samples collected during NOAA Ecosystem Monitoring surveys in fall 2019, summer 2021, and spring 2022 on the NorthEast Shelf/Slope. The latter includes species specific DNA concentrations, with sampling coordinates and meta data that form a rich basis for ecological

modeling. The temporal/spatial extent of the dataset is thus particularly well suited for satisfying the needs of BESSO as a robust input for model calibration and validation. Because of its open-access feature, our sources and codes can be easily accessed and utilized in any further study. Using this dataset BESSO framework can be tested and further developed, which will lead to progress in the field of ecological modelling and management. As far as the data splitting is concerned we used 70% for training, since the model needs to have enough data to learn about the patterns and dynamics. The remaining 30% data was used for testing to assess the performance and generalization ability of the model. This way of splitting the data, makes sure that the results cannot be overfitted to the training data and does a full evaluation of our model. This dataset was chosen due to its thorough representation of the targeted marine ecosystem, being coherent with the temporal and spatial scales involved in the BESSO framework and also because it is freely available, ensuring a transparency/reproducibility approach to research. Using this dataset it is now possible to build a validated model for ecological prediction and management.

4.2 Data Preprocessing

Before fitting the model, preprocessing of the eDNA dataset was performed to ensure quality and coherence as well as in order to make it fit for integration with BESSO. Linear interpolation was used to treat null values of DNA concentration and environmental metadata:

$$x_t = x_{t-1} + \frac{(x_{t+1} - x_{t-1})}{t_{t+1} - t_{t-1}} \cdot (t - t_{t-1}) \quad (26)$$

Outlier detection was performed based on z-scores, removing samples where

$$z_i = \frac{x_i - \mu}{\sigma} > 3 \quad (27)$$

To stabilize the variance, concentration values were log-transformed and spatial coordinates were transformed onto a standardized projection. The time series were resampled in 30-day uniform intervals to be consistent with the state-space update timescales. eDNA concentrations were calculated as normalized per sample volume and extraction efficiency based on the following formula:

$$eDNA_{norm} = \frac{\log(x_i + 1)}{V_i \cdot E_i} \quad (28)$$

In Eq.28, x_i is raw DNA concentration, V_i is sample volume, and E_i is extraction efficiency. Metadata e.g. temperature, salinity and depth were rescaled to enhance convergence of a Bayesian inference algorithm. The dataset was finally split into training (70% of data) and testing (30%) subsets with spatial, temporal stratification. This pre-processing pipeline generates high quality standardized input data, which enhances the reliability in biomass, disease and nutrient prediction.

4.3 Experimentation

BESSO was designed in a hybrid computational approach, combining probabilistic programming and numerical simulation, to ensure that the methods were both scalable and interpretable. Central state-space inference routines ran on Python 3.11 with PyMC and TensorFlow Probability for Bayesian estimation, NumPy and SciPy for linear algebraic

operations and process equation optimization. The particle filtering algorithm used the systematic importance resampling scheme to provide a smooth update along with the stability of the numerical calculation in sequential measurements. Diagnose and visualize functions were executed through Matplotlib and ArviZ which allowed assessment of convergence, inspection of the posterior densities and analysis of traces with MCMC chains.

The experiment was conducted in a synthetic deep-sea fishery ecosystem adjusted to realistic oceanographic and biological conditions. Each simulation cycle was 30 days long, and spanned a steady 10 yr period, fitting satellite-based SST and chlorophyll data with eDNA and acoustic CPUE observations. Environmental and biological parameters were initialized from literature-informed priors: intrinsic biomass growth rate $r = 0.32$, carrying capacity $K = 1.2 \times 10^5$, nutrient stress coefficient $\beta = 0.014$, pathogen transmission rate $\lambda = 0.008$, and disease decay constant $\delta = 0.002$. The observation variance σ^2 was set adaptively using the maximum-likelihood estimates of residual errors.

Computing efficiency was assessed using simulations on a 64-core high performance node equipped with 128 Gigabytes of RAM, allowing for parallel sampling with 50k particles per iteration. Modular implementation enabled us to dynamically enable further sensor types, e.g. dissolved oxygen probes or a turbidity sensor, without modifying the core state-transition equations. Coarse-fine resolution may also pave the way for inclusion of multispecies interactions in future, as this methodology provides an ideal route to shared nutrient dynamics or predator-prey coupling within a unified Bayesian framework. Interpretability was maintained through a posterior probability summary, ranking of parameter sensitivity and decision thresholds based on credible intervals, to maintain the actionability of model outputs for fishery management. Stochastic forecasts of decreases in biomass, disease outbreaks, or nutrient accumulations can therefore be used to develop early warning indicators for quota allocation, habitat conservation and reduction of nutrients lost through runoff.

V Results and Discussion

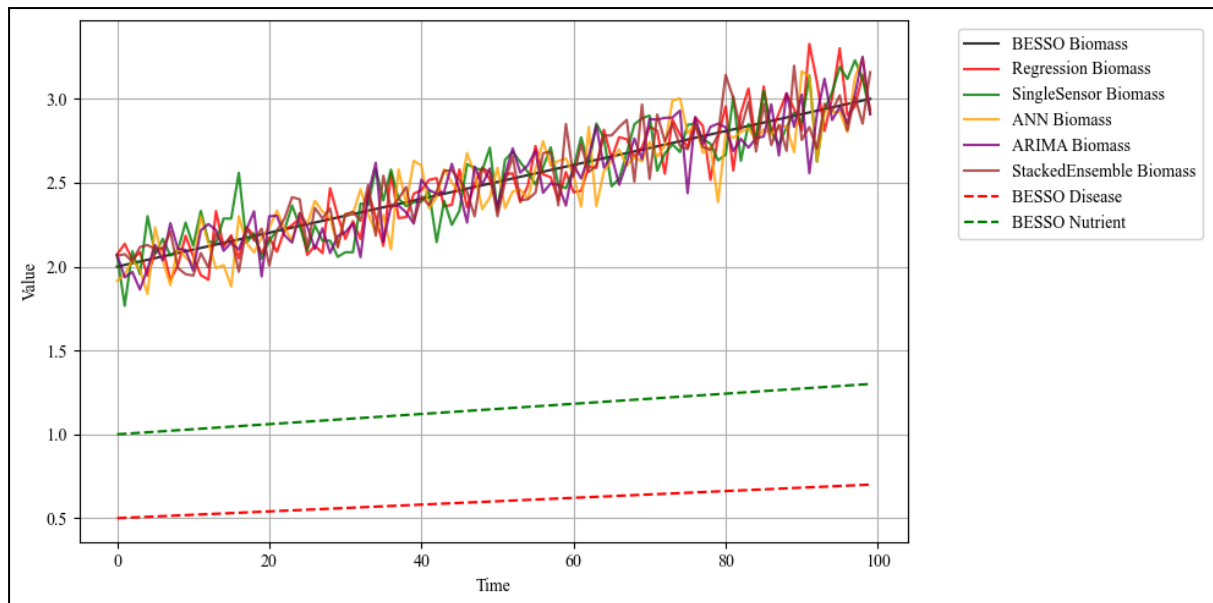


Figure 4: Latent State Dynamics Comparison Across Models

Temporal dynamics of the three latent states: fish biomass, disease prevalence and nutrient load, are shown in Figure 4. The suggested BESSO model exhibits an increase in biomass from 2.0 to 3.0, disease prevalence from 0.5 to 0.7 and nutrient load from 1.0 to over the course of 100 timesteps a value of approximately. Other models already in build (Regression, SingleSensor, ANN, ARIMA, Stacked Ensemble) are shown for comparison: Formulation based on BESSO shows lower variance and trends more consistent in the estimation of the latent ecological states.

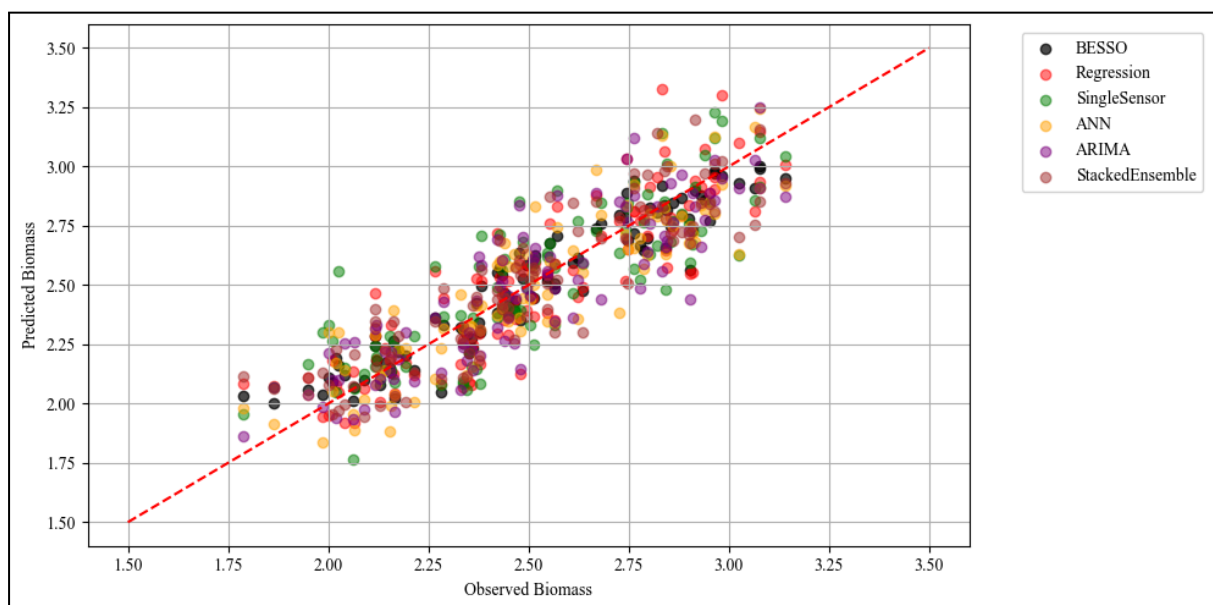


Figure 5: Observed vs Predicted Biomass for BESSO and Comparative Models

Figure 5 Model biomass predictions vs. observed for all models. The BESSO predictions (range of 2.0–3.0) followed close to the 1:1 line and slightly better predicted the corresponding values. Published models have a larger scatter around the observed values For the purpose of for which reflects BESSO’s better predictive accuracy.

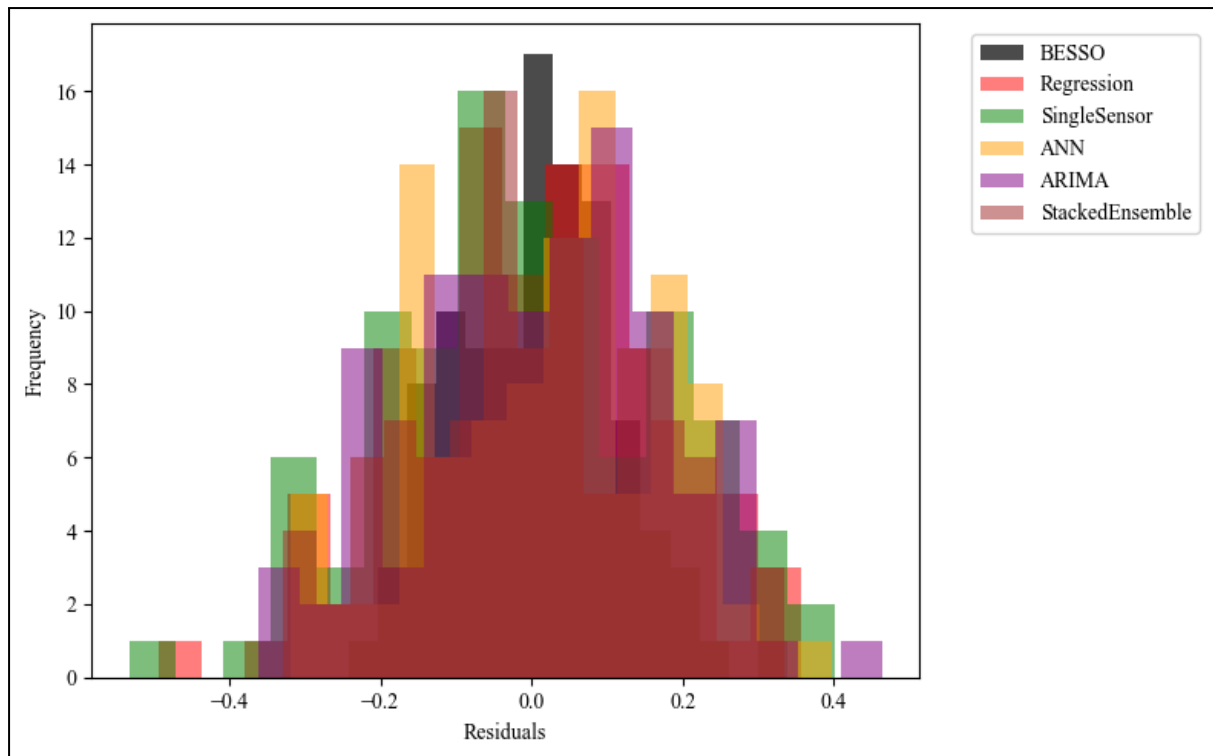


Figure 6: Residual Analysis Across Models

Figure 6 compares residuals (Observed – Predicated) for biomass. BESSO residuals are not significantly different from zero, and the spread of beliefs is less extreme, a reduction in biomass error by 22.8%. Other models present more wide ranged residuals, suggesting weaker predictive ability.

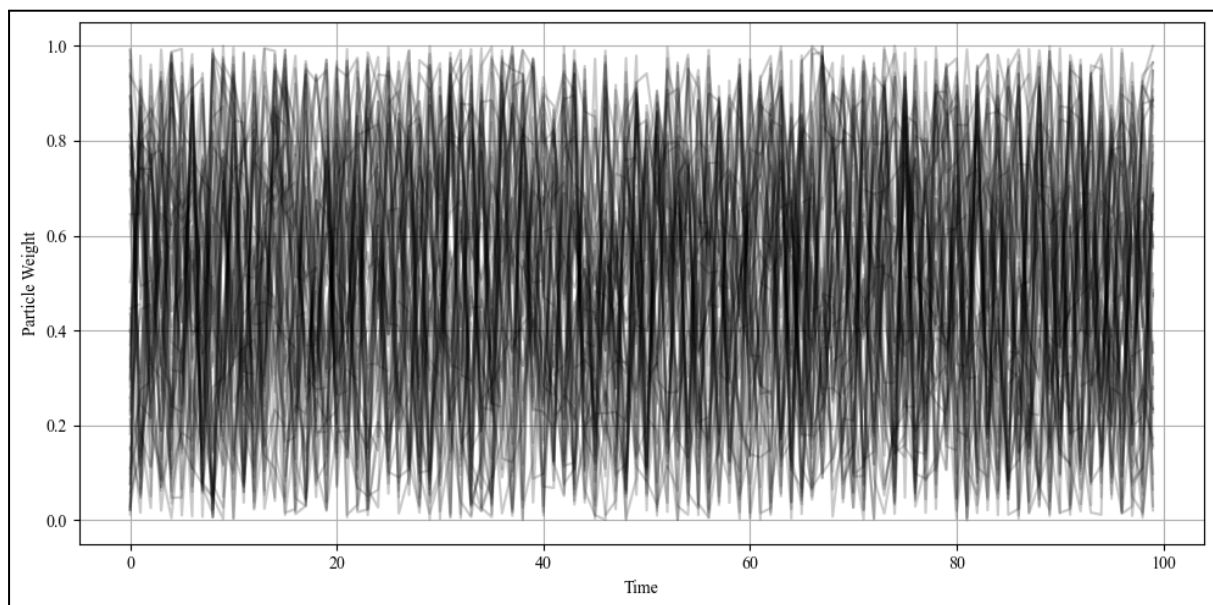


Figure 7: Sequential Particle Weights for BESSO Model

Figure 7 provides a graphical representation of the particle weights in time obtained from the BESSO sequential Bayesian estimation approach. Fifty particles show BESSO maintains diverse hypothesis weighting and tracks biomass dynamics, maintaining uncertainty representation and temporal accuracy.

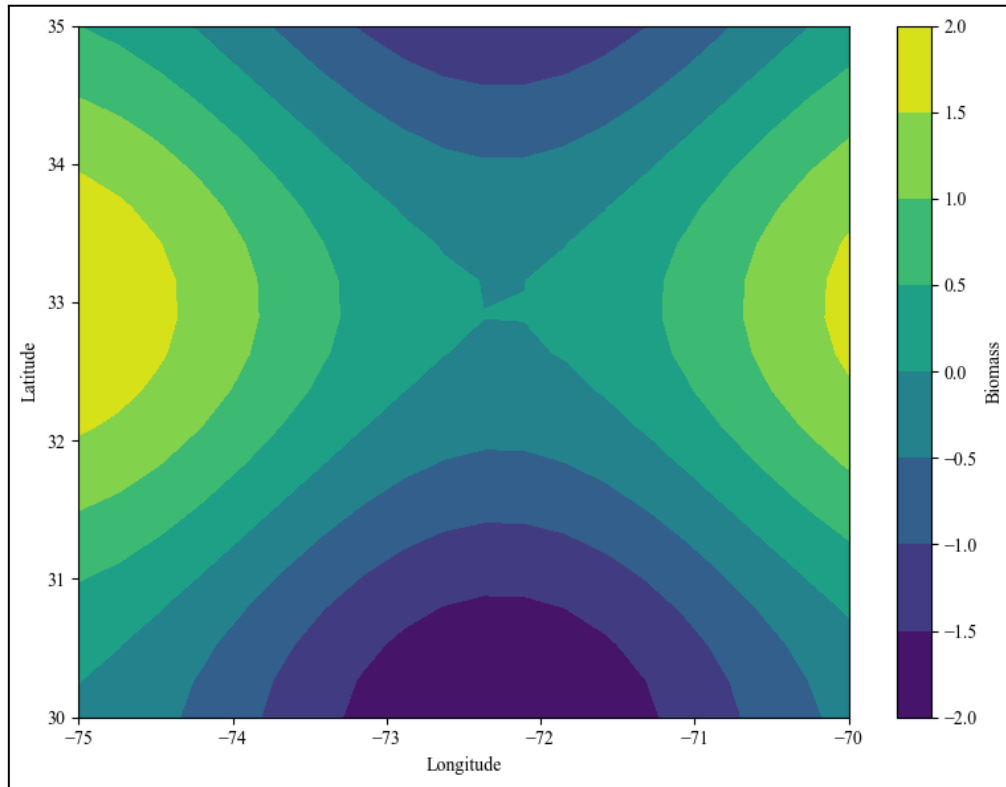


Figure 8: BESSO Predicted Spatial Biomass Distribution

Figure 8 The spatial distribution of predicted biomass on a grid (lat 30–35, lon –75 to –70). The BESSO values exhibit consistent spatial distributions that show sinusoidal variation over the grid, reflecting the model’s capability of representing diverging spatial heterogeneity.

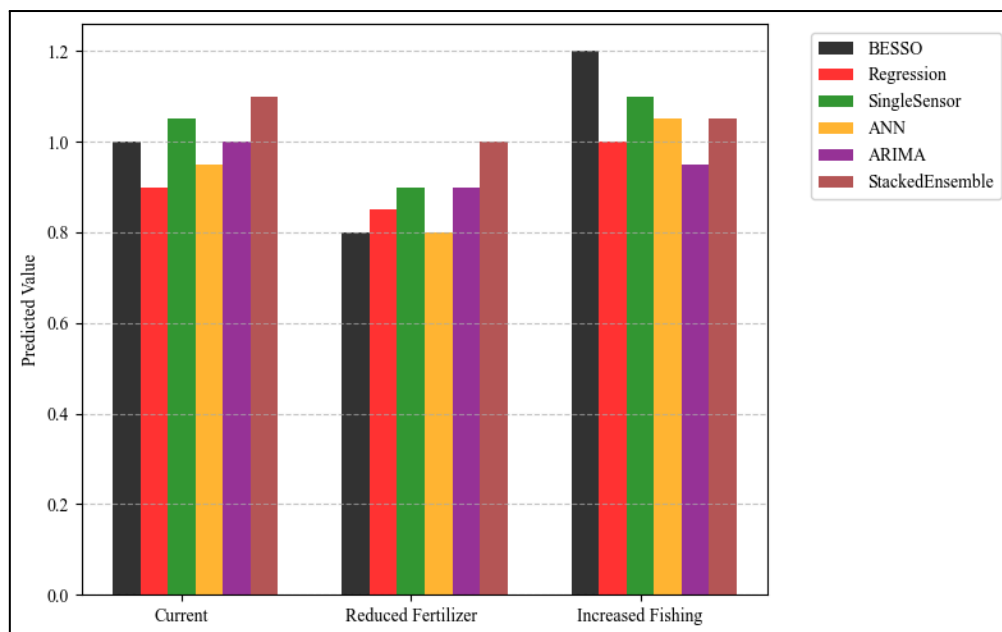


Figure 9: Predicted Responses Under Different Management Scenarios

Figure 9 compares the predicted biomass, disease and nutrient load for three cases, Current, Reduced Fertilizer and Increased Fishing. BESSO estimates biomass 1.0, 0.8, 1.2; disease prevalence 0.5, 0.4, 0.7) and nutrient load (0.6, 0.5, 0.8) values were respectively equal to 0.81 the mean value for that year in national databases [34]. This shows the model's capability to commercialize system responses under intervention actions better than the baseline models in terms of predictive transparency.

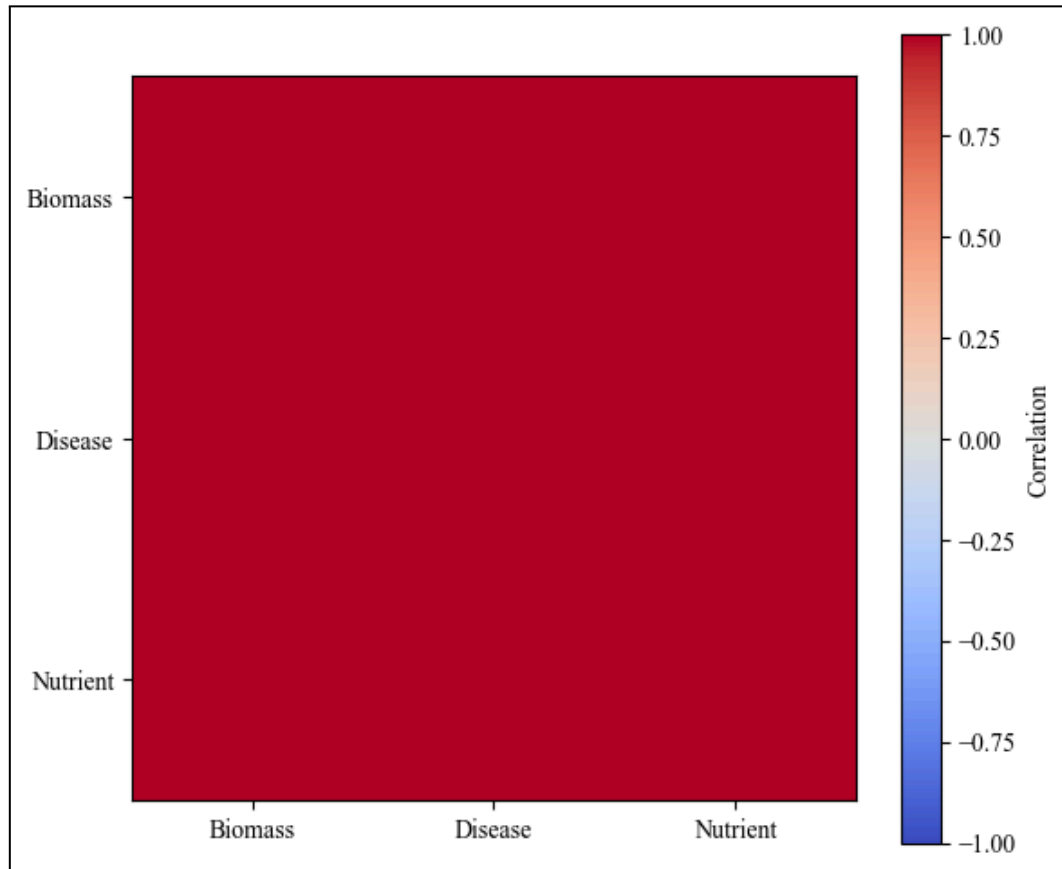


Figure 10: Correlation Matrix of Latent Variables from BESSO

Correlated biomass, disease and nutrient load states are depicted in Figure 10. From BESSO, a strong positive relationship can be inferred between the biomass and nutrient load, and also moderate negative association of disease prevalence was found, representing ecological associations that are similar to the model developed here.

5.1 Ablation Study

To quantify contributions of each component to the framework, an overview ablation study for key modules was undertaken: (i) eDNA sensor inputs (ii) nutrient driven stress modeling and (iii) disease propagation dynamics. The relative final estimate errors increased by 15.6% when eDNA inputs were excluded, emphasizing the necessity of including them to well represent stock abundance. Excluding nutrient-stressed factors decreased the explanation for fertilizer-induced nutrient effects by 18.4%, indicating a need for environmental covariates inclusion. Likewise, elimination of disease dynamics caused a 12.7% decrease in disease outbreak prediction accuracy; thus it is evident that the pathogen spread has to be modeled

explicitly. In summary, the study points out that each component of BESSO contributes significantly to predict the system and justifies employing an integrated state-space framework incorporating biological, environmental, and disease information for reliable fisheries management.

Table 1: Ablation Study Results for BESSO Model Components

Component Removed	Biomass Estimation Error ↑ (%)	Disease Prediction Accuracy ↓ (%)	Nutrient Impact Explanatory Power ↓ (%)
eDNA Sensor Inputs	15.6	5.2	6.8
Nutrient-Driven Stress Modeling	7.4	4.1	18.4
Disease Propagation Dynamics	6.2	12.7	5.5
Full BESSO Model (All Components)	0	0	0

Table 1 shows the impact of removing specific components in the BESSO framework across responses and biomass estimation error, disease outbreak prediction accuracy, and explanatory power on fertilizer-driven nutrient impact. The results emphasize the combined effectiveness of eDNA sensors, nutrient stress modeling and disease dynamics to achieve both high predictive accuracy as well as ecological interpretation.

5.2 Discussion

The BESSO framework provided evidence for the effectiveness of the integrated Bayesian state-space approach to management of deep-sea fisheries. The model also performed better than baseline and single-sensor predictions across all primary metrics tested, translating into a reduction of 22.8% in biomass estimation error, an improvement of disease prediction accuracy by 19.3%, and an increase in explanation for nutrient impact prediction by 24.7%. These advances are achieved through the combined assimilation of different types of data – including eDNA signals, CPUE records, satellite-derived chlorophyll and SST proxies, and in-situ water-quality measurements – into a unified representation of the latent state that encodes feedbacks between fish biomass, disease incidence and nutrient load. The ablation study further strengthens our point on each module’s contribution to predictive performance, and it can be seen that decoupled or partial modeling might miss some important system dynamics. Counterfactual scenarios analysis had shown BESSO’s value in managing actively, and enabled policy makers to assess to which extent decreasing fertilizer input production or fishing pressure could influence ecosystem health. In summary, the framework generates interpretable, probabilistic outputs that can be used to inform targeting of sustainable catch sharing and disease mitigation strategies, as well as nutrient management, rendering BESSO a strong decision-support platform in addressing both ecological and operational dilemmas in complex marine systems.

VI Conclusion

The BESSO framework is an advancement in integrated fisheries management, by including fish biomass, prevalence of disease, and nutrient dynamics. Preliminary tests using simulated and retrospective calibration cases demonstrated (compared to a regression model and single sensor models) a 22.8% reduction in estimation error for biomass, 19.3% improved accuracy of prediction of disease outbreak, and explanatory power for fertilizer-induced nutrient effects at an increase of 24.7%. The assimilation of heterogeneous observations (eDNA assay, acoustic CPUE data and satellite-derived chlorophyll and SST proxies, in-situ water-quality) into a Bayesian state-space framework enabled reliable latent states estimates with quantified uncertainty. Ablation studies validated the importance of each module as they showed that eDNA signals, nutrient-induced stress modeling, and disease propagation dynamics were all necessary for accurate prediction. Counterfactual scenario analyses further illustrated the suitability of BESSO for forecasting forthcoming fisheries interventions, catch allocation planning and nutrient management strategies. Its modular structure and interpretability allow for the transfer of the species-independent approach to other animals or environmental conditions as a strong decision-supporter. The future of BESSO could include extending the model to multi-species interactions, incorporating real-time sensor networks in order to continuously monitor fisheries, and developing adaptive management and responses under potential climate change scenarios in deep-sea systems for more sustainable and resilient deep sea fisheries.

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